

ActInf GuestStream 061.1 ~ Shagor Rahman: "Myth of objectivity and the

GuestStream | Shagor Rahman | 1:29:52

hello and welcome everyone to active inference Institute it is October 20th 2023 and we're here in active guest stream 61.1 with shagor Ramen so today we'll be having a presentation myth of objectivity and the origin of symbols followed by a discussion so thank you very much for joining looking forward to the talk thank you very much s for coordinating this um and and thanks yeah for all the the work that you and the active in active inference Institute does um also wanted to briefly thank um Carl fren and Paul B Badcock for um helping me with this paper and kind of making it a reality so this is um the title of the paper here so it's called myth of objectivity in the origin of symbols um and it's in Frontiers and sociology um and so just get right into it as um as Daniel said and do you hopefully a brief overview and we can do a discussion at at the tail end this um and so I guess briefly the thesis that I'm trying to argue for uh is that and you know some of these words will need clarification um particularly in in the context we use him um but it is that the belief in Universal and kind of a codified egalitarian um social norms or in other words morality this essentially underwrites broadly our use of symbols um so what is that kind of mean functionally in terms of what other

disciplines use um or how they would think of our use of symbols um well there's cultural Niche um and this you can think of as you know our ability to engage and cultural Evolution um from Joseph enrik and others who have highlighted that as an area of an important area of Humanity's distinction another one is social reality and this is um more in the psychological realm um but you know how we use things like money um you know how we um obey someone with a crown on their head um so just things that are socially constructed um realities are socially constructed as opposed to you know existing in more of a physical sense that other um non-human animals do and of course this is gets to our symbolic capacity so this is you know our symbolic Consciousness our um ability to kind of think deeply in the past to plan deeply in the future um and to engage with kind of any number of um Concepts as well as people uh and so the claim we can divide up this claim into kind of three areas um the first is our kind of modern social dynamics and social bias and so you know how this is the sociology side um so you know how our economic systems um look how our Civic uh and political institutions look um and kind of building on um that so I think once we have the structure on hopefully I convince you of um a way we look at um Humanity's

uh structure of their social reality and
know their kind of symbolic reality um
then you know the question is well how
did we get there you know how did we
become um the kind of you know primate
that points fingers and feels guilt as
you can kind of see in that picture um
and for this we can look to you know two
areas so one is you know non-human
social animals so pites um you know
cations and another animals that engage
in social behavior of a similar nature
um so this is distinct right from
usocial insects from Mano ants and bees
um but rather this is you know chimps um
you know
bonobos marasa monkeys and and some
others that have been studied um
specifically you know by researchers um
seeking to examine Humanity social norms
and how it differs from you know primat
and other animals and and kind of lastly
is is the are of of anthropology um so
whatever we're able to you know
determine about our you know the social
environments of our ancestors um you
know what can that tell us about how we
got here and I'll make some kind of
specific
claims in
regards
um yeah and so so I set out my goal and
essentially my goal is to kind of tackle
that first part and to you know persuade
folks um you know on the call
here that
our social system takes on a particular
structure and that structure is a

essentially a symbolic self and I use
that in in the sense of
carlong meaning that we inhabit um in
addition to our own identities we
conceive of ourselves our
self-conception uh is attached
to a number of other um broader cultural
identities and these these could be a
this could be um you know your country
this could be your company um this could
be your community you got to think of
you those those 5c so country company um
Clan Etc um and so that's essentially
where that star is that that's where I'd
like to get to um by the end of today
and I'll also get into some of the other
areas as well um and so to to explain
briefly what I mean by the myth of of
objectivity uh so first of all um by
myth I I mean to say that we have
it's not that we've obviously identified
some kind of underlying structure with
you know reality or with our social
systems um that our symbols represent
right so think here of um Plato's forms
that you know there's um you know a
number of chairs out there but there's a
kind of an ideal form of chair um you
know that's not the claim I'm making one
way the one way or the other um so
instead what I'm saying is Well during
the um past scene we were just like
another primate um trying to solve
primate problems um and the way we did
that or one way we did that um
effectively um was through this was
through what constituted essentially a
myth was essentially

um establishing um you know a common reference frame um for behavior and I'll get into that term in a bit in a second and the second um piece of this the objectivity so that's to mean that it's not simply um it's not simply a claim that you know our ancestors um you know listen to stories and you know believed in myth and believed in religion um but that our myths that were powerful um were practical um Were Meant to solve um social problems and do so in um an objective way so be essentially more scientific um so one of the claims that I'm making is that our ancestors kind of similar to how we are today um or sorry similar to how some of us are um are interested were interested um in identifying um actual real world consequences um in a symbolic manner so having some sort of objective truth um and kind of you know believing that an objective truth exists um so kind of right and wrong exist in an objective way um could in bad exist in the sense of um you know this can kill me this this can't kill me um and objective in the sense of you know this is part of my in group and this is outside of it all right so roughly how did we get there um so more specific um on the uh anthropological side um um I'm G to look at things like shth um so that this is our most innate social bias um which is important because um for a few reasons but one is that um Society these social social units even outside

of the human Spectrum um some
researchers have used um things such as
um social bias to
analyze um the structure of non-human
animal social units um so essentially in
group preferences um social biases in
that respect um are ways that we can
understand how other animals how chimps
and bonobos interact with one another
how they establish friends and um and
distinguish between VES and essentially
how they exist in these um you know
normative um you know normative social
units and hence our use of shth speaks
to kind of a unique way that Humanity um
expanded upon our social biases and so
although it's you know a Crux now um you
know it's it's often seen as
problematic um there's a certain
contingence of folk that c should as um
a necessary step in our Evolution um so
I adopt that view and um see that as a
an important kind of factor in our road
to to morality right um and so within
that kind of shth environment
um um which is essentially we can think
of as still non symbolic um because shth
um as um I should probably should
probably explain what should with this
um so this is essentially accent bias um
so we're kind of taken from um the Old
Testament but it represents uh our kind
of innate tendency um to associate and
to establish an instinctive kind of
common ground with folks who sound like
us um and so we you know automatically
trust folks um you know who grew up near
us who have a similar accent um and the

further out we get um often times um the more foreign some can see not always the case but kind of instinctually and you know prior to the acquisition of symbolic identity we'll use sh will will typically um you know practice shth as a way to to um identify and so you know assuming that this that this practice of shib withth you know goes kind of far back into our past um then you know how do we transition out of that um to something more symbolic you know how do we you know start using morality how do we start using language um well here our uh my suggestion is that we use one of or one way to think of it is using um what's called a speech act and so this comes out of the um the field of the philosophy of language John Surlly and Austin a bit earlier than that um but there's speech acts called performative and constitutive utterances and these are essentially acts that um immediately change uh the environment in some way and so um if I suddenly tell you you know that you're fired um or that um you know um um you know there's a there's a war um across the seas right now um that has immediate consequences even though it was just a speed check so before any violence happens before um someone is now kind of crowned a king and starts implementing physical policies just by speaking you've changed something um and so in this kind of examples my kind of stylized example is that um well someone committed

um some kind of Act of deviation they violated some Norm um and so some vocalization was emitted that um labeled that person so this person should have been an ingroup member this person kind of should have been aligned with our in group's Norms but they violated in some way and it could have been you know they violated food sharing they they violated um you know some of Unwritten contract about Cooperative breeding like how we that we should protect um you know infants um um and it could be done a number of different normal violations but labeling that person um you know as a deviant would constitute kind of a minimal threshold of a speech act um that could be read as symbolic and so the the starting off point here actually is um or one of the important starting off points here actually um is what is itself s and so um you know symbols are are one thing and um we can probably identify that and and have some discussion but we have some kind of clear idea of what the self is um sorry what a symbol is it turns out however that you know defining the self is a bit um it's it's not as confirmed and locked in um as at least I would have expected kind of when I when I started this um this project and so typically agents are um defined we get we'll get to that in a second they're they're U they're defined as along their person um but it's it's

not clear where um say a cell starts
where um an organ starts um you know
where a person starts and where Society
is um and so what we'll get out here is
um using some Frameworks we'll see how
we can understand bias perspective
shared go goals reference frames um
and the next topic which is um the free
energy principle see how we can use
those to start understanding um you know
how self and how
identity how how we can think of um the
self and identity in the context of um
you know our creation of symbols
right and so yeah roughly I mentioned
some of these already but um what we're
going to look at is non-human sociality
um we look at you know religion myth
narrative this is you know of course a
very instinctual very ancient um
practices um that defines Humanity um
we'll get a social bias as I mentioned
um and then ancient sh and most
importantly you know this is the active
inference Institute um and so the
framework that I'll U be positioning
this within um is the free energy
principle and more specifically Act of
inference
um
yeah just just one more over um so again
fren principle um we'll go over the
ideas of identity uh and how we can
start to think about this subject and
how that is evolving um in the light of
these new Frameworks of the fre in
principal active inference and some
other researchers um who are sympathetic

um to these
views uh and then the the weird journey
and I'll explain that's an acronym so
I'll explain what that acronym is uh in
a bit but essentially for now it's just
you know Western Society it's so Western
Society is zone for being particularly
analytical um and as a result of that
one of the consequences of that is um
most recently and this is you know
around the past Century so around 20th
century a lot
of um disciplines emerged um that did
something um that had a common approach
um which was an attempt to um you know
reduce uncertainty um identify agents um
and you know um provide kind of
explanatory Frameworks in the field of
um that first one post structural this
is in the field of linguistics the
second one of course is economics um and
the third one is behaviorism in the
field of psychology and so we'll just
kind of go through those and see what we
can learn from um these paradigms um
when we try when I try to essentially
you know convince and and um know
discuss
Humanity as um you know an agent that
occupies a symbolic
self yeah so in addition to of going the
route that um I mentioned so far we'll
also discuss what this means for moral
philosophy um and you of course this is
we we'll just
um kind of dip our toes in we won't go
too far into moral philosophy but I
think the implications are kind of

immediately um clear in some areas um it brings some interesting ideas from the past of you know what is a bit more relevant um what's a bit less relevant um whether you know effective altruism and things like that are kind of consistent with this view or kind of whether they're um at odds um and the last piece if we have time just because uh I know there's a similar stream on um Cooperative communication so um I'll kind of give my view on how this um kind of aligns with Cooperative communication and um communication viewed as kind of an ecological framework which you know both of those were um purportedly done within the active influence framework so yeah and so uh to start things off um the free energy principle so just kind give a brief explanation so is the active inference Institute um so there's obviously you know a great kind of wealth of uh information um that explains the Paradigm and how it's useful and some of the terms but essentially it's a framework that essential that um attempts to identify um a number of formalisms um that describe the science of self-organization or autopoiesis and it represents uh agents things objects um everything as essentially statistical models of their environment uh and so more specifically agents need to possess a um that minimum uh they need to possess

a generative model of their environment
in which they
can understand in a sense or infer
the processes of the world so you know
how does an agent you know a creature
navigate his environment how does an agent
you know find its way back home they
have the they themselves represented as can be
represented as models of their
environment so we can see that as as
uh these agents encoding basic
beliefs

um and um so one way to look at and this
isn't part of the formalism that's why
it's that way um but I I heard Chris Fields
say it's all I'll say it here which is
that one way to think of it is you know
agents are essentially all scientists
they have beliefs about the world um
they act on the world in a certain sense
um and those beliefs get updated so
everyone has a running hypothesis of
what they should next do who they are
and how they should maintain themselves
their environment and so at a basic
level you know everything that exists in
the world has to engage in in some sort
of

Science and this

uh occurs at various scales um so this
is at cells at organs um you know agents
people um or animals of you will
societies and and culture and that kind
of last area of course is where um our
focus is here

today and this is a large size as well but
um one of the takeaways from the free
Ander principle is that a foundational

element is the search for oneself and so
it's not um a given that we understand
precisely who we are um in the world and
it kind of extends to who your friends
are um and who your enemies are and um
and um but this surprise imization uh
can be thought of as this um
Socratic search for self you know
meaning you know Socratic Socrates
victim to to know thyself um there's
something ancient there and not just um
you know for moral
philosophy yeah and and one other point
to make is that um although belief uh in
the FY principal context is um a
technical term it's it's refers to
um basian statistics it does actually
come out of um a use at a higher level
so Ed Thomas spay was a preacher and he
was specifically talking about religious
Faith um and it was a formal and he made
a formalization for how we he could
think about um his religious beliefs and
so that's ultimately what led to um
basian statistics um and
so one attempt or one um kind of
takeaway I I I have is that we can think
of due to um principal scale and variant
nature we can think of um beliefs that's
we typically use the term the religious
context and and another context um we
can you know try to formalize these
within the free energ principle um not
as a given um you know we could get
things wrong um but it's at least
conceivable that they're contained
within
that all

right the next area so in addition to
um the free energy principle there's of
course other work that's been done to
understand identity understand self um
and whether that exists kind of at the
bounds of the skin or whether that exist
you within social units um and so one
interesting

um U one interesting piece that's been
done um is by Michel Lan who focuses on
bioelectricity at the cellular level and
how cells communicate

and there he actually used Game Theory
which is a paradigm typically used in um
in economics but he demonstrated how uh
cells if we can think if we can model
them not just as kind of individuated um
little agents trying to maximize their
own interest if we can set mod of them
as potentially inhabiting um a wider
self we can see how cooperation would
emerge um in a way that

um more competitive paradigms more
reductive paradigms would die and so
immediately even before we get to how
Humanity acts with other humans um
whether we do that cooperatively whether
we use morality with use other Norms
um the same problem exists you know at
the level of the

cell um because and the second piece is
once we get to um once we get to
chimpanzees if we're over in the the
upper right hand corner um what we uh
what we understand there is um for Mark
Moffett Sanz determine their group size
by social intimacy and so this means
that their group size essentially cannot

exceed generally does not exceed um what
any individual chimp is able to
recognize so they basically have to re
so each person each member of the group
has to recognize everyone within that as
opposed to you know as opposed to us and
humans um which the already use has
existed for you know far into our past
we've been in Anonymous social units
right um and so here um what I did was
before I get to um further down the
line um what I did was take what I I
took another concept of Michael Evans
which is this light cone so um the light
cone essentially is one way of
representing
identity and so the the scope of each
agent goals um can be kind of captured
in this idea of like cone and so in in
the area of a chimpanzee there their
goals are associated with others in
their that that cohabitate in their
social unit um and they're necessarily
subjective and so they're necessarily um
based on their own
perspective
right and so another piece of social
identity another way to look at it um
and so
there's a debate whether or not um we
can model animals as being in societies
and so this isn't a given um and so
there's you been arguments between um
you know dawkins and um Edward Wilson
for instance that you know claim that e
side kind of claiming that you can you
either can um think of animals as um
existing in Social units and those

operate similar to cells or they can't
one way around that um on this point
number three researchers found that you
can model them actually as just
possessing ingroup
favoritism so if instead of kind of
worrying about specifically where the
boundary is from our point of view from
um you know an outsider's point of view
if you inste model it as well each
individual agent um so like I said Each
one um
maintains uh an intimate recognition of
others within their in group um
following that they show preference
refences to those inside so this could
be as simple as when there's a fight
when there's an attack from an outside
group you'll prefer um those within your
ingroup versus out group um it could be
food sharing it could be hunting and
things like that but it doesn't have to
be extensive cooperation it could be
simple um preferences it could be
avoiding spending more time around your
in group versus um outside your outside
group and so this is one of the ways in
which um social identity
um you know exists at this at this
level and so another important piece is
um which Judith
burkart um in K's measured which was
implicit Norms uh and subjective Norms
that that are exhibited by these chimpan
and
so as an example if chimp saw um an in
group infant being harmed um they would
experience um a visceral reaction um

they would have heightened Razzes they would pay attention um and so this kind of suggests that there are norms that chimpanzees and other animals um and other animals exhibit and some animals even exhibit a more extensive um more extensive Norms such as marasa monkeys um but chimps are on the lower they still they're on the lower end of that Spectrum but they still exhibit norms and so one way that um Judith Bart distinguished between um Prime immorality and human morality um was to say that we have a top- down third party perspective um and so this is where I get the goofy um graphic where if there's a triangle on top of um each of these each of one of our ancestors heads um is because our ability to engage in objective morality um can be represented by this shift between um perpendicular uh perspective perspective where I'm looking at those around me and and I'm imagining them looking back at me versus I'm taking a top down perspective uh and evaluating my actions and the actions of others in the context of some more transcendental um perspective and so this and that object that view of objective morality um versus subjective um social norms we'll say was also brought up by Beck off and Pierce um they did claim that morality exists in other animals but they were using Morality In A broader sense non

symbolic

way um and they make a lot of good points with this is and they go across wolves they go across um crustations you know lobsters Etc um how these animals engage in essentially behaviors that suggest Norms but also exhibit behaviors that suggest emotions that are underlying that we typically think as underlying um Norms of

morality and so the last piece I'll mention is that

um morality for humans is typically um is typically symbolic is pretty strongly symbolic um and the work of um Shin gr and others um will typically or have found that in

order to identify an action or a behavior as being a moral constituting um an immoral Behavior it'll typically need

to

um violate some principle or some code so continuing um continuing on

the idea of what establish as a subject um you know how do we first identify what an individual is and what can this tell us about how we manage something like a symbolic self and how does a symbolic self then become to be constituted by U you know by um a belief in Morality or what is good for the overall self versus the individual the work of Mark sus who used active inference and the freeangel principle um to understand the function of

emotions and what his one conclusion he had is that emotions is the the spot of of Consciousness um and that should be the direction of consciousness research um so I won't go that far but one point that he makes that's relevant here is that the um the phenomenon of emotion is relevant specifically in helping manage categorically distinct needs um so distinguishing between um you know pain and outside pain and hunger which one's more important um do I go to sleep or do I have food managing these various you know different substance systems um is the function of emotion and can start to help start to help us understand the you know the boundaries of the self and so if we go back to um thinking about how mik 11 um looked at an agent how these things Coes together we can see how emotions fits nicely into that picture um because it helps essentially harmonize these VAR disparate systems into a person right this so get this gets at honic veilance this is kind of pleasure versus unpleasure so this is an important Dimension there's a few Dimensions um that researches have identified you know with regards to emotions so you know how do we study emotions in in a rigorous fashion veilance of course is a important measure there so kind of pleasure versus unpleasure you can think of it as pain um but typically it's this understood specifically as pleasure

versus unpleasure um and it's in that sense that complex organisms start to move and and help navigate the world and on so social creatures um how they navigate the world you can see too how it could help us conceive ourselves as within the symbolic self and so um some research on the idea as a symbolic self so what I'm hoping here is that my idea here is that although this was brought up by previous psychologist at earlier times um I think it's clear that you know thinking of subjectivity and thinking of uh symbols in the context of active reference and you know minimizing uncertainty um and a more expansive view of the self um the Frameworks that that I list here can become more operationalized um so Carl Young found that you know this the self as symbol or the cosmic man was a motif that existed in numerous cultures um from you know Hinduism to the ancient Greeks um you know to ancient China there was is this conin theme of the symbolic man and so each person's part from that view um you know from a religious persective from theth mythological perspective um everyone was see conceived as part of a common person so this helped um establish essentially a common reference frame for behavior um so this is and and similarly Victor Frankle Locos of therapy um who's a holocaust Survivor and also a psychologist who emphasize the need for humans to find meaning in groups and ultimately

find a meaning um in lives and our well-being is was largely determined by that um yeah just kind of broadly um our sociological background so in a sense we do a number of things um that we kind of take for granted but that are all essentially symbolic and that all you represent our um you know ability to inhabit not just our own identities um but we attach ourselves with cultural with cultures uh that are in some cases extinct and in the past with Futures that we kind of have youf to um understand so I do want to and discussions think I'll get to the to the final thesis with um but I I would like to kind of briefly go over um the weird Journey that I have here um and so uh I think this demonstrates how a couple things um and but analyzing um Western Sciences so Western science uh paradigms in the field of economics and and language and behaviorism can help um bring some of the the points I'm making to light and so we are the acronym that I mentioned earlier this is um Joseph enri um it's Western educated IND industri highs rich and Democratic uh as you can see so essentially what he found was a lot of uh scientific studies relied on college students um they were typically college students in in wealthy countries and they were well of course well educated because they're in college um and this caused a lot of research to have certain biases on that if you start

including other um other societies that are less um weird um hunter gatherers for instance Etc you get different results so he came up with this acronym um but then he he doubled down on it and in a more recent book he explained that um weird Psychology was the reason for the Industrial Revolution um the reason for the scientific um mindedness that that we exhibit um and Etc and so this and we differ and the consequences of that is we possess an analytical mind so you know think of you know a scientific oriented mind um um you think of someone who's bookish um versus more Galt um thinking which is present in more feudal and cast and clan-based um societies and institutions um so thisal thinking had you know has led more recently and it was a um a discussion a couple days ago on on using narrative Theory and economics um so that's firmly in this um you know in the message I'm trying to um try to overview here but of course in neoclassical economics is an example of a more analytically oriented but something that is essentially reductive and structuralism and behaviorism follow along the same kind of heels and so what these three U what these three paradigms have in common is that they attempt to explain a complex phenomenon by um reducing uncertainty although minimizing uncertainty is important within the F principle it's also important to um

understand that it's that it's minimized
and that minimization happens um you
know dynamically within systems and is
part of the you know adaptive process um
and essentially
how we get the systems that we do and
how they are Dynamic how they change
over time um how we have preferences um
Etc and

so

so one question is well if if these um
you know if these paradigms were overly
reductive uh and don't do a great job of
of you know

explaining the systems that they purport
to then why do we use them one reason is
that they are great null hypothesis um

so

um you know as we'll see a lot of
economists like tasan neoclassical
economics a lot of post structuralists
like to bash on structuralists um and
that kind of goes on and and you know
there's good reasons for that because
you know sometimes when things are bad
they're they're um good at help
elucidating important Concepts that are
true okay and the second reason that I I
bring it up today is that um they reveal
moral conception at the heart um of how
Humanity deviates how our social systems
deviate from these predictions um and a
third reason is that the hund gatherers
were also analytical um so although they
differed from us in important ways one
thing they shared um from us was an
analytically oriented mind so that means
that the clan-based and and the feudal

institutions that Henrik was talking about this happened only past the Agricultural Revolution so this was 10,000 years ago um when most societies essentially shifted their normative structure became more stationary and as a result of that adopted the these more GTI views so

one way to look at um our trajectory two symbols was well our our um you know our ancestral past um was filled with people trying to do science using symbols they trying to categorize um plants and animals hunting methods so anything they used for cultural Evolution and most importantly they try to classify one another they try to type one another um I'll do one example

um from that and then I'll I'll move it over to a discussion so there's a few um areas that uh I wasn't able to get to but I'm have to do that as a follow-up and then I think most importantly it be great to get some questions in um but yes there's the idea of transcendental signification so this is one example um where the poststructuralist the philosopher Derrida and P had a similar critique um of what was known as structuralism so for some background Cloud Levi

Strauss around 1960s um maybe a little bit before that ideology known structuralism which attempts to view cultures view language and some other um institutions that he examined as corresponding to an

underlying and stationary
structure and so it's removing the
variability from different cultures from
different myths and believing that
there's a certain structure to
this and DED and other's critique of
this was well essentially we're drawing
arbitrary lines and culture isn't so
easily um reduced to a specific culture
um there's time there's history there's
of cultural variation um and one example
is um EV trying to draw a hard line
between nature and
culture when really that those things
are continuous in a different cultural a
different natural environment you might
say um you'll have a different culture
um and the culture will feed off of that
natural environment and so as darus
there's nothing outside the text so
everything is a matter of perspective um
in that sense so the question then
becomes well if uh if this if that
static structure that CL Levi stous um
purported existed then does do cultures
have any structure and of course DED and
others still did believe that there is
room for a structural science right and
so the structural science we're looking
at today of course is the free and
principal on active inference um and so
what is the kind of core components what
is that um I didn't go over the concept
um today but where is that you know
Markov um blanket or
boundary that we're minimizing
uncertainty within um and that's where
the transcendental signification um idea

comes in and this where said well
there's typically especially in Western
societies um there's typically a concept
that you just don't that you can't argue
against that's just something accepted
um and from there that the constant
daisy chain of Concepts the constant
flow of Concepts can stop essentially
and so you think of any um at any given
time it took me a minute to think about
what our transcendental signification is
today um you know because there's a lot
of fluid concept and there's a lot of
arguments um but one examp example is
freedom so basically at least so lan so
if someone brings
up in a political context or certain
other context that this will infringe
upon Freedom um then it's very hard to
argue against that and that's because
it's it goes without saying that um you
know although there could be in some
context infringements on freedom it's
generally understood that you know
freedom is a good thing it's a positive
thing um
and
any kind of disagreements from it will
have to begin with that
rightness and so although D was critical
of
U Western society's use of transational
significations and um you know that's
one modern example but of course in
others are you know the belief in God um
democracy other things that are just
taken for granted that um the culture
should abide

by um and although he was critical of
these or our use of them without
transcendental signification is
difficult to imagine how a culture would
be maintained I think it's also
difficult to imagine how you know
language would um remain stable that's
because there needs to be some agreement
um there needs to be essentially some
common ground in order for communication
to happen um that's because of course
language involves adopting a whole bunch
of

Norms um but also our daily lives that
are kind of empowered by language and
symbols um need some sort of um of
stopping point need some sort of um
ground to say well you know this this
isn't a good thing or this is a good
thing we shall kind of Orient ourselves
to that

um and so this is one representation of
um why moral conceptions are um kind of
necessary for our systems um and you can
see that in language

and you can see that I think in um in
economics and the field of social trust
as well um that I can go into that kind
of later

time

um yeah the last thing is um this
relates as similar things happen with
the motion but I'll stop there actually
and um you know folks want to do kind of
a followup we can we can do that
but that's want to turn on see if
there's any uh questions at the moment
so hey

Daniel wow well awesome a lot of uh pieces arising so for those who are watching live they can write questions in live chat and I wrote down a few things as well so just while we get started I mean what brought you to bring in active

inference and we explored the role in the coming questions but what what brought you to this were you studying theories of cognition and looking for what or how did it come to be this way so I guess techic I took the high road I took the high road to inference so I

um I

was studying economics and I thought to myself well we have economics has a behavioral framework um it might not be great but it's it's related to um utility it's related to welfare it should actually be a pretty good starting point for understanding morality so essentially it could be a utilitarian framework um except that we have prices so we should be able to clearly at least propose you know a set of um you know policies and and set of behaviors or a set of or an essentially a normative theory that you can score Behavior against to say whether it's moral or immoral um so that became very hard because it turns out that um the the neoclassical Paradigm doesn't allow that because it um it fixes individuals uh and their and their preferences um and so uh I then kind of turned to um some

other moral paradigms that that used
actually prisoners dilemma um and others
that brought in Morality but then yeah
then then I saw the um this the
Scientific American or the wired article
about Carl friston and him talking about
the free engry principle um yeah and it
seemed like okay there's there's an
option here there's um you know a way
to you know use math like economists
love to do um but kind of in a way that
still allows for um you know true
subjectivity and kind of true you know
variation um without kind of reducing
that all the way but yeah in a nutshell

I
guess I I I really like this never never
even heard it that way like math with a
true subjectivity it's truly about the
subject and what is subjective to
them a lot of ways to go from there um
okay you brought up
shth and
syntax and shith at least in its
original um biblical style it had to do
with how the word was said so it it was
truly a question of of the mor ology of
the word it's not like it was an unknown
word it was how it was said not the
underlying term um and then you seem to
suggest
that we were in a new phase or mode of
semantic
communication that there would be
something past or different than the
shth so what what was that pointing to
or what do you see there yeah sure I um
I definitely ran through the word sh I

didn't get to the um to the um
anthropological side yeah so so
essentially the what I'm referring to
here is um I guess he had two things so
one is Emily Cohen um has a paper called
the case for accents um Mark mofin has a
has a book called human swarm and he
kind of Echoes that that same argument
there um and the last piece that I'll
mention is cat kathern Kinsler um has a
book called How you say it um and
they're all about shith um the first two
were about making the argument that um
what we call shith and and what I mean
I'm using it here in the context of it's
just accent bias right so that in that
case the um the eedes you know they or
one of them killed the other one because
they pronounced shth incorrectly they
said cth instead of shth I think um but
the point is that um you know from the
second where we're in utero um we listen
before we see anything um we listen to
our cons specifics before we see
them
and there's a lot of evidence that we
show preferences immediately for people
who
sound like the people that we
heard initially sound um so we establish
these a kind of orbit around our Social
Circle
um using that familiarity in sound um
and a lot of kind of racism and a lot of
other social bias can actually be
deconstructed into relating either more
to um shth or the accents are actually a
bigger driver than we realize and we

consciously realize um and so Emily
Cohen um uh and Mark
Moffett they use that as a way to
suggest
what if we had accents kind of before we
had
words um and how would that how would
that
look and the idea here is that well
other animals have
vocalizations um you know dog bark um
chimpanzees um make vocal vocalized
sounds verit monkeys they do these up
and Below calls so they have in some
cases sophistic sophisticated
vocalizations obviously parrots have
very sophisticated vocalizations only
next to us
um and so would
we would there be a circumstance where
uh our answers are could have used that
um to identify with one another um in a
more cultural evolutionary perspective
so why do we do it as as babies why do
we do it as as children um well the
argument well that probably is
indicative of something that's more
evolutionary ancient that has more has
more legs to it um and the reason why
that's important is because you know we
we started off as a m primate we somehow
um you know did you know defeated the
chimps and the monops in terms of
resources or whatever
um and you know how did we start to kind
of cooperate at scale um and you know
there's been some discussion of that but
introducing ship with clo is a lot of

gaps for instance in Joseph henrik's um
um you know the story of our success um
or in his first book on car Revolution
he goes through how

[Music]

we BEC how we became a species that
learned from one another um and he said
he admits that majority of this happened
when we live close close together but if
you think about it well we must have
learned from each other while we were
still in these sparsely commit
communities we we transitioned from
these sparely communities to these
higher dense environments but um it'd be
very difficult to imagine a densely
populated um you know area of chimpanzee
because there would be a lot of fights
so how do we kind of overcome that and
Sh is one example by kind of extending
our our our networks but yeah it's a
long way to answer to your
question interesting so kind of the
historical and
developmental Primacy of accent and then
the kind of linguistic and
symbolic signification that we have as a
scalable
technology that's why seemingly an
open-ended number of people can have a
word that they might all pronounce
differently um or uh simple abstract
simple like the triangle that you showed
that that might mean something different
to everybody but it's like this and and
that's where you kind of invoked the
platonic concept which is like the
word separate from any accent like does

the logos have an
accent exactly exactly yeah yeah I was
try to I spent like a lot of time on the
the the image because uh I want to be
careful because it's not that there's
one um there's one light that we're all
accessing right it's not that there's
one perspective true perspective as far
as we know that we're all that you know
that Daniel's accessing when when you
talk to me and that I'm accessing when I
talk to you um we just have kind of we
both
have uh a technically Transcendent
perspective that sufficiently overlaps
to the extent that our communication now
is is what you might call productive
right to the thing that we can um
properly communicate um we're
co-constructing that little triangle
blimp um which is kind of yeah modifi
modification of of Michael Evans kind of
light yeah accent encoding in the
production of speech accent
decoding in the recognition of speech
and then that's what
allows for example token based language
models to be in our cognitive
ecosystem and get some meanings or you
know allow them to be projected meaning
upon but but cool
um Okay so so where does active
inference come into this for you like in
the paper and in the presentation you
mentioned a few places but you brought
in so many
historical Frameworks and different
directions so just in your own work

leading up to here or from where you
take it like what do you see active
inference in the free energy principle
doing in that
line I I I see it
as so I I I see it as
operationalizing um the Y notion of you
know symbol symbol AS self um and I I
think that does uh a lot and
um it does a lot because it gets at
that uh foundational one of the
foundational principles which is um
seeking self so
uh it's not that you know we're born and
you know some children decide you know
I'm not going to speak um you know I'm
not I'm not super interested in in this
or I I I I find that I belong in this
family they give me food and I'm just
going to stay here
forever those kind of things don't
happen um and so what I what it does for
me is it's it frames the motivation um
for symbolic communication um because if
if it's conceived of as an extension of
your identity um if this cultural unit
is in some
ways more foundational to human identity
than even um a chimp social unit is for
its identity um which I think it is then
you know it no longer becomes an option
so it kind of accelerates the epistemic
uh ontology that I think the free
principle um
underwrites meeting kind of ser for self
seeking information information gain so
I think you could and I didn't but I
think we can rephrase this as morality

as information processing um so in in
that context in the context of I'm
a person I'm essentially a concept
within a broader symbolic social reality
um and it starts to I start to minimize
uncertainty by um Computing cells by
finding you know the the blanket that
I'm behind
um then morality in that case becomes a
signal to what those identities are and
where I belong so in that case um
someone was deviant okay this person is
evil then that's that's a big deal um
let me pay attention to this um
someone um you know yelled at me when I
crossed the street because I was
crossing
the street at a wrong time that's going
to stick in my mind because they've
they've pegged me as something right so
that was the use of morality as creating
something um and so yeah so this is one
of the the reasons morality is I think
so Loom so large I think in in our minds
um and I think is is important for for
that check but yeah yeah you also have
insight into you know these Concepts so
yeah if there's the words I should be
using please please stop me
here yeah I think it's very interesting
to
apply active inference to the yion or to
the symbolic Journey to the search for
meaning and and all of this so if we
were just talking about like visual
perception pixels on the screen or
photons on the
retina a sophisticated cognitive entity

would do structure learning and propose latent causes like oh maybe it's not just a million red pixels moving from left to right maybe it's a red car and so then there's essential or the symbolic car so that would be like structure learning and World model again also for the kind of entities that we are it's like a selfin world model because one of the most important things is agency and understanding well what are the movements that I have agency over which is why we're able to kind of ignore is secades because we know or or suppress proprioception associated with our body being in the position that we expect it to be because those bodily features are under our control and so we're we're able to basically calibrate them out not need to be distracted or attend to them and then that allows us to be sensitized to changes that are not caused by us and so it's like what is the eye what is this symbolic and embodied strange Loop that's causing all my thoughts my thoughts and so it's like that can be framed very clearly in a uncertainty reduction structure learning external state or hidden State framing

[Music]

and only um through kind of a lot of um modification and and coercion can that be cast in like a microeconomic reward maximization framework which just leaves more

um to be done than before the
application and ACM is like well no this
is at least
a uncertainty reducing way
to
approach that question of self and
potentially multiscale self um
Discovery yeah like in
um yeah one of the the points I was
trying to make with um neoc classical
like similar to what Dar did in um in
structuralism was
well you because this this is such a
this is a general equilibrium kind of I
think of it as like a thin um framework
meaning it um it reduces the economy
down to um essentially get no movement
because all the resources are allocated
and there's really nothing nothing left
to do um
but the what's interesting is is when
people identify where that fails so you
know classical market failures um it's
the most interesting side of um you know
classical economics it's it's how you
get um pollution it's how you get
technical spillovers that have you know
positive reaction um I think an
interesting one is in the concept of
cultural niches uh social trust and
so there the and in also the um and how
firms Co see how firms don't exist so
you'll have social trust at the level of
um level of a country you'll have social
trust at the level of the firm um
and within that that minimizes
uncertainty for the employees it
minimizes uncertainty for the consumers

because they know the product they're getting um and I think we can also think of it as minimizing uncertainty for um you think of someone who swell and they're not in the marketplace yet they're just learning things um so what's you know how do they participate um how can we think of them participating in the market economy other than just a um kind of a longtime consumer until they start um participating well they have they have a series of of of symbols and they have a series of ways of reducing their uncertainty um and so this is you know unions this is um universities this is you know companies that maybe they're not thinking in companies per se but they're thinking of um you know a business person or a scientist or you know they have these similar things that can be used for there tell another question was you described this kind of perspective swap that we engage in and you described it as there was a a perpendicular component to what what was that social setting that was perpendicular and what was perpendicular to what yeah yeah and so so here um you know so so so here I guess I'm kind of the on the edge of my not knowledge but what what I was the perpendicular would be uh so typical um primary relations but

also um but also a lot of our social relations are are intimate and so that's perpendicular meaning in the sense of um you know for these are the people that you that you you know these are interactions that you have that are kind of based on um your subjective um perceptions and meaning those meaning those interactions that we can kind of carve out of a symbolic reaction so for for I guess for for people it's it's difficult to do because as soon as you engage with someone um you know you're thinking about what you should be doing and and and you know there's a bunch of unconscious things of okay I should put you know I should shake their hand or you know I should um you know if I'm I should kiss their cheek or um you know I should take my shoes off now so there's obviously kind of a top down implicit um symbolic um occurrences happening however you know for non-human animals for for primates um you know for dos Etc um that world is you know it does have it does have history um you know it does have it does have memory but the perspective is still is still subjective it is still you know a subjective memory um there there's no kind of Above The Fray uh objective top down um sense for which they can coordinate their their actions and behaviors and their kind of goals um and so I on on one hand I don't think it's saying too much because I I think it's

basically just describing um you it's kind of visualizing how symbols are different than um symbolic interactions are different than um um non-symbolic interactions on the other hand I think it's interesting because uh

I think it can

help it can help us imagine the context and maybe model the context for how we first started it right how we we first started to do it um which is that we were in an environment where um both environment both in a in a um in an actual environment like in in the place but also our kind of cognition was at a place where we could start to imagine our Behavior as if a third person were looking at it um and so that yeah that's essentially what I was what I was getting

at cool all right I'm going to go to some questions from live chat so if you if you have any thoughts on them please feel free otherwise of course however you want to see them okay upcycle Club wrote how can tokenization improve the performance of active inference agents that use tensor networks as generative models for sequential data and then just followed up by defining tokenization is the process of breaking down a text into smaller units called tokens which can be words subwords characters or symbols do you have any thoughts on

this repeat the question yeah yeah how can tokenization improve the performance of active inference agents that use tensor networks as generative models for

sequential

data you know I was just interested in
in the question itself but yeah I have
no I have no thoughts but other than I
can't wait to you know to examine the
the wording of that
question and maybe have an opinion one
day yeah appreciate appreciate whoever
asked that whoever asked I thinking that
I would have insights into that question
so

indeed it's on stream we'll come back to
it okay Dave asks a question there's a
few background pieces

but it will end with a question mark
from what I see okay Whitehead in
process in reality presents a framework
that allows a kind of chaining across
local spatial relations that allows a
group of observers to attribute
objective locations to objects in a
shared space a group of

Minds can apparently share Concepts
somewhat contrary to what we'd expect
from a strict avoidance of the f y of
misplaced concreteness it'd be nice to
speak meaningfully of shared
Concepts while avoiding the fallacy of
misplaced concreteness and also avoid
Whitehead's rather unsatisfactory
Reliance on Eternal objects Concepts
residing in the mind of God could the
trick be made to work by positing
chaining across locally I and thou first
first and second person mappings where
you and I take into account the partial
discrepancies between what you and I
mean by a given term or a given illusion

while we retain and reuse the structural
commonalities between your and my usage
of the
term yeah I mean I this this is
certainly much more in my um in my
wheelhouse than the in the first one um
and yeah I mean I I start my paper
talking about wiad so I I certainly
deserve a question that involves it um I
I I think precisely and and I I
think that that's kind of getting at
what um you know the way I
present kind of our Discovery in
language we um
approaching the idea of of you know the
F the fallacy of misplace concreteness
like um we believe in these platonic
forms it's this Western bias it's this
thing that Plato did that was wrong that
we continue to do with you know
classical economics but but uh even
though it's wrong it's it motivates us
to create language because by believing
something um by believing in these forms
in a sense um we end up reifying we end
up creating things that that do exist so
I I guess the way I see it first of all
is that um the myth of objectivities is
that
well by wanting to believe that there is
a right way to speak by wanting to
believe that um there's a right
thing there's a right thing I mean by
the right thing um we end up kind of
bringing that about um
uh where I'm curious to kind of pars his
question out is at the tail end where he
talks about you know the diff the

distance between I guess the two
subjects I and you um and kind of how we
kind of inch closer together I guess
might simp framework I'm thinking well
if if both me and you believe that
you know we're speaking more or less the
same language um you know we're both
talking about active inference like even
even at at lower levels then we can
navigate a closer and closer space and
establish something like a reference
Frame where things where words have common
meanings Etc um we can establish some
common ground um and I think yeah so I
think on one hand you can think of it as
this

if I understood correctly towards the
end we can think of it as this kind of
closing the gap between our two
perspectives um but I think you also
have to it's also necessary that we're
both conceiving of this
um you know I don't want to say Godlike
because we were trying to avoid white
head's use of Godlike but it Godlike is
is useful because um it if we just think
of that there's a a that there's this
fallacy and there's this um perspective
that can see the true form um so we in a
sense we need this kind of Mythology I
think um or we at least needed it in the
past um to kind of get at something like
language

forms awesome all out a few a few notes
on that and ask another question so in
thinking about
these appeals or
abductive hypotheses about hidden causes

it's like wow what is causing all of these horses I mean there's small ones and big ones and they have different patterns and some have four legs some have three legs one you know had this other number of legs one time so it's like what could be causing all of these sounds sights smells and there's basically two extreme structural hypotheses one of them is like the single like there's the that's the platonic horse it's like I don't even know what it is it's very different than any of the horses but it's just somehow is the structural cause it's the formal cause of all these Horses The Other Extreme structural hypothesis is like horses don't exist everything causes every like there's just sort of like an infinite component model now complexity is always found in between the one and the infinite and so that's where we actually do develop more well articulated accounts and what really does cause a horse to be a certain way in its accent in its real embodiment but we can see that kind of um monist or absolutist appeal as being like as you put a a good null hypothesis like potentially one of the best starting null hypothesis not going too deep into value judgments but I mean why not just start with saying maybe there's one thing and then we'll keep adding latent causes as we go but then to confuse the appeal to a single cause

with the actuality of the single cause
is kind of this challenging dance
because if you take it as real then it's
real as
real and then to hold it at a distance
though but it's hard to hold something
at a distance when life and everything
is on the
line um and then one other piece what
you said about like well the the the
Godlike appeals of of um some of these
like whiteheads formal approaches and
you seem to express somewhat of an
ambivalence about this kind of an
appeal um and that reminded me of the
quote from Hamlet what a piece of work
is a man how Noble in reason how
infinite in faculty in form and moving
how Express and admirable in action how
like an angel in apprehension how like a
god action and inference in Hamlet early
ACM

days okay I'll ask a question from the
chat Muse equal
wrote isn't this indicative of learning
an intuition being entropic
postobservation we disassemble the
observed construct into segment frames
of historical time through
interaction prior knowledge and
memory where where is learning where is
symbolic learning in what ways is
symbolic learning anticipatory real time
or
retrospective I guess this kind um you
know what you were saying with you know
forms as kind of causes um of hidden
States and kind of how do you um you

know update you know update beliefs like
once you have the lece like how
complicated is it to um you know to
actually update them
um trying to think of an example that
that we went through
um but I yeah it's
a and now now I'm just thinking of of my
daughter right because she's she's
learning how to speak right so she's
clearly symbolic learning um
and you know kids are like much more
comfortable um with variation um you
know with with denture pre um I'm not
comfortable but they they exist in much
more enty than um than than than we are
um so they can they can navigate the
space I think probably better than we
can conceive that they that we can or
that even one can um because they're in
it so often um and how do you kind of
build symbols well I
guess um Okay so
where I came from from um symbolic
learning was um hoffet and and in
Sanders's
um um book on analogies I'm forgetting
surface is in essence um and so their
their claim is that
um which I think aligns pretty well with
active inference um but I'm curious to
hear more feedback on it um and and
their claim is
that um we learned primarily through
metaphor enemy um and so it's through I
think through analogy they said so surf
they look at we look at surface
differences rather than essential

differences and essential would be more
form platonic form um look at more of
sural differences so um um you know my
you know my dad is to me there's a dad
towards someone else um you know I'm I'm
crossing the street so I I I I suppose
learning um I'm trying to think of some
good examples but um you
know we we
build we
build structures
of uh symbols um using kind of like
against like so you know this is
um
like you know this is an action of
sitting in in one chair um I sat I saw
someone sitting on a bench outside
um you know I saw someone sitting on the
ground um so
it's producing I guess if you a
generative model of um you know in this
context in this situation this is
happening um and these are the um you
know these are the relevant features of
the environment um and this
is then the object within that
environment um here's something going on
here and I'm bringing in and I'm using
that as as a prior to establish
something um and then at some
point and then at some point we we start
getting Concepts and then once you have
your initial concept because that takes
some time once you have your initial
concept then you compare Concepts to
Concepts in this kind of metaphorical
fashion I didn't give a good example of
metaphor but um we start to move outward

that way but yeah I guess that's the
best that's the best I have for you at
the
moment yeah this role of metaphor is
very interesting and something lightly
speculated upon but not quite brought
into active inference generative
modeling yet but metaphors is like well
it's like the highest point on the
building or it's like the center of the
circle or it's like what's inside or
outside of the circle so those would
those kinds of metaphors which can be
bootstrapped or grounded in in the
immediacy
of our Developmental and
evolutionary
history can then be brought
into purely cognitive or symbolic spaces
so we could talk about a clash of
symbols or about friction or inertia and
grasping all of these
um
tangible examples that are kind of out
there but then we bring them in here and
I I think that may have something to do
with what Dave was discussing with the I
and thou so it's not that's kind of the
Martin boober it's not just I it if we
were talking to um an inanimate object
or an object that we didn't think had a
selfin world
model then it'd be like ranting to you
know this puppet or just like kind of
just but but under no expectation that
it has a mind of its own but then the I
thou or in active inference what's
sometimes called like thinking through

other minds or thinking with other Minds
where it's kind of like well I'm
modeling my own speech in relation
implicitly or explicitly to what they'll
respond to right and and and knowing
that they have that kind of recursive um
activity as well that is purely
cognitive it's it's lifted from the
Accenture it's brought into the
structural and the
symbolic um but it's always realized by
systems with
accents and and I I I think just the way
that
you introduced and and
highlighted um that
more
um direct experience
of like how somebody is saying that not
what they say it's easy for us now to
focus on the semantics of the what which
are so
overwhelming but that
almost obscures the kind of
bootloader that actually got us to the
developed social systems which as you
pointed out clearly some things have
changed such that we don't require that
direct intimacy
yeah and so that that's a a complex
tangle of of cognitive and cultural
phenomena it's not like only being
ascribed to to the ability to do
metaphor or think through other minds or
use symbolic um logic and learning but
all of these are are coming together in
this kind of unified mystery
yeah yeah yeah and um yeah maybe

understand um that earlier point so the
the frisen article on um singing
um singing on from the same H sheet um
or you think of it as you know with with
language you're engaging in
these um you know fast-paced Dances um
with people you know exchanging um you
know exchanging
beliefs in anticipation of um you know
your in interior beliefs um and you're Co
co-constructing something um yeah I the
only thing I want to say is uh when you
ask earlier where I felt Act of
inference was was useful I guess in a
nutshell it
can it can
better understand the Dynamics between
cooperation competition um you know
outside and inside
um and yeah that gets us some of the
tension with how we think of
communication whether it's Cooperative
or whether it's kind of just ecological
um
but it it encourages a
um it encourages that Dynamic where
we're we're in um like if you know if
we're saying something and maybe I I'll
disag maybe you'll disagree slightly
with me on a point um but we're so we're
still here trying to be cooperative um
so we're trying to say things that we
both understand um and sharing that kind
of same cultural me space or attempting
to and at the same time
um
proposing either iteratives or
modifications of those beliefs so that

I'm able to persuade you of my point
um when we get out whether communication
is kind of Cooperative um to the extent
that it's Cooperative to it's dance
versus it's kind of um
contentious then you can start
to tease out those know I don't know
those those factors
okay short and interesting question in
live chat upcycle Club
writes what do you think about
petroglyphs petroglyphs are rock
carvings made by pecking directly on
Rock surfaces using a stone chisel and a
Hammerstone what do you think about rock
inscription Rock inscription shely I'm
big fan no U well I imagine you might be
getting at is um uh is that uh
so is that symbolic you know probably if
human does it if um maybe he's getting
it or she's getting it if
a you know if if a bird has done it is
that symbolic um yeah I don't know I'm
curious but yeah I'm a big fan of of
sculpture that kind of reminds me of the
oam andus poem like I was here and all
of that entails like there was an Empire
here now of course in the poem it's all
degraded
and so that's kind of the media and the
message but the inscription on the stone
is like I was here I chose to do this or
even
if sorry what oh almost like graffiti or
something like like why why are we why
are we doing this why are we okay I get
it yeah somebody somebody who uses that
graffiti tag did that we could have

different stories they could have
different stories about whether they
have free will or all these different
like secondary things but it's like the
fact of the matter is on the rock it was
inscribed so I I I say yes so I mean I I
get I get that now so um what I see
there is um you know what I see there
is that
is you know that's soul right that
that's uh in in in the sense of um um
you know something like like Whitman
would would describe it is that that is
um we we do that because we have a soul
ademy not that in a um not necessarily
in a religious context but um as in you
can exist uh you know daniiel Freeman
can exist as a conceptual identity
outside of space and time um and the way
you do that is that you you leave a mark
you know you leave your legacy and um
you know um you know etc etc but yeah so
we we exist as these symbolic selves
both um as part of these cultural units
but in some sense outside of it um and
so yeah do you write your name on a wall
do you um you know do you start a
company um you know do you want to be
um you know famous do you want to be on
a movie for whatever reason um yeah this
is that that is an extension of identity
um yes so if that's what that's getting
at then yeah
um cool well in sort of
closing where are you going to take this
line after the Frontiers
paper
and this what's the next Frontier or the

next

Journey AI yes so that that that I don't know so yeah I'm open to ideas but um yeah yeah I was I was obsessed with this idea for a while um and I think I've nearly gotten it I think I'm kind of like 95% the way there or maybe 90% um and I think this discussion has helped me but yeah I'm just kind of curious to see um where it might go but yeah I mean I'd love to see it formalized love to see it taken seriously but um yeah have no specific uh has no specifics but cool it was a great with you I appreciate the setting us up and I appreciate the dialogue thank you I really appreciate hearing the comprehensiv and just the relatability of the Curiosity and how many things are brought together in your paper and in this discussion so see you around till next time all right take care thanks everyone bye